

Transmissibility of adenovirus-induced adiposity in a chicken model.



BACKGROUND: We previously reported that human adenovirus Ad-36 induces adiposity and paradoxically lower levels of serum cholesterol (CHOL) and triglycerides (TG) in animals. OBJECTIVE: To evaluate the transmissibility of Ad-36 and Ad-36 induced adiposity using a chicken model. DESIGN: Experiment 1--four chickens were housed (two per cage) and one from each cage was inoculated with Ad-36. Duration of presence of Ad-36 DNA in the blood of all chickens was monitored. Experiment 2--two groups of chickens were intranasally inoculated with Ad-36 (infected donors, I-D) or media (control donors, C-D). Blood drawn 36 h later from I-D and C-D groups was inoculated into wing veins of recipient chickens (infected receivers, I-R, and control receivers, C-R, respectively). On sacrifice, 5 weeks post-inoculation, blood was drawn, body weight noted and visceral fat was separated and weighed. RESULTS: Experiment 1--Ad-36 DNA appeared in the blood of the inoculated chickens and that of uninoculated chickens (cage mates) within 12 h of inoculation and the viral DNA persisted up to 25 days in the blood. Experiment 2--compared with C-D, visceral and total body fat were significantly greater and CHOL significantly lower for the I-D and I-R. TG were significantly lower for the I-D. Ad-36 was isolated from 12 out of 16 blood samples of the I-D that were used for inoculating I-R chickens. Ad-36 DNA was present in the blood and the adipose tissue of the I-D and I-R but not in the skeletal muscles of animals selected randomly for testing. CONCLUSION: As seen in experiment 1, Ad-36 infection can be transmitted horizontally from an infected chicken to another chicken sharing the cage. Additionally, experiment 2 demonstrated blood-borne transmission of Ad-36-induced adiposity in chickens. Transmissibility of Ad-36-induced adiposity in chicken model raises serious concerns about such a possibility in humans that needs further investigation.